An external evaluation of the DRfold2 single-sequence RNA
structure transformer on 11 post-cutoff X-ray crystals, with four reproducibility findings from a head-to-head with AlphaFold 3

Vincent Le Duigou*

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Abstract

We report an external evaluation of DRfold2 [1] (a single-sequence-input RNA structure transformer, training cutoff end-of-2023) on 11 X-ray crystals deposited after that cutoff (57–101 nt, December 2024 to March 2026), plus the 2021-deposited CPEB3 ribozyme as an in-distribution reference. On the 9 blind monomeric crystals, DRfold2 reaches a mean C1′ Kabsch RMSD of **4.21 Å** (median 3.99 Å, range 2.06–7.72 Å). On the 2 blind multi-chain crystals (a tetrameric GTP aptamer and a dimeric dopamine aptamer with ligand-bound interface), DRfold2 fails (6.00 and 16.82 Å) consistently with a monomer-input / multimer-context mismatch. We additionally surface three reproducibility hazards from a side comparison with AlphaFold 3 (Server, default) [3] on a 5-target subset: (i) submitting multiple RNA sequences in a single multi-chain Server job catastrophically degrades each chain's prediction (R1108: pLDDT 56 and 6.5 Å as single-chain vs. pLDDT 25 and 25.6 Å as one of seven chains), an unannounced framing artifact; (ii) the apparent ranking of two predictors is dominated by training-data cutoff asymmetry — DRfold2 (end-2023) and AF3 (Sept 2021) cannot be fairly compared on pre-AF3-cutoff structures, but we have no RNA-only crystals in the cell where DRfold2 saw and AF3 did not; (iii) on the 5 targets where we have both, AF3's per-residue pLDDT is rank-correlated with crystal error (Spearman ρ down to -0.58, $p < 10^{-6}$) on 4/5 targets, providing a useful abstention signal even where the prediction's mean error is high; (iv) DRfold2, which exposes no explicit confidence head, has an implicit per-residue uncertainty signal in the standard deviation across its 80-model ensemble that is positively rank-correlated with the crystal error on **12/12** targets (**10/12** raw and **6/12** Bonferroni-significant at $\alpha = 0.05/12$, with the strongest case at $\rho = +0.66, p \approx 10^{-11}$). All code, AF3 ZIPs, and crystal comparisons are at `https://github.com/Vince-Anduril/rna3d-calibration`.

1 Introduction

DRfold2 [1] and AlphaFold 3 [3] are the two most-cited RNA 3D-structure predictors of the 2024–2025 cycle. They occupy opposite ends of the input regime: AF3 builds its own MSA inside its black-box Server pipeline; DRfold2 is a single-sequence language-model architecture (the RNA Composite Language Model, RCLM) that takes one RNA string and returns one ensemble of 80 models from which an energy-based selection step emits a final structure. Direct benchmarks between the two are sparse and confound several variables: training-data exposure (the PDB cutoffs differ by more than two years), submission framing (AF3's Server interface accepts multi-chain jobs that behave

*Albert School Madrid; `vleduigou@c2tclinic.com`. The analysis pipeline, code, and a manuscript draft were generated with substantial assistance from Anthropic's Claude (Opus 4.7) operating in a structured agent loop under VLD's direction (see Methods and Acknowledgments). VLD reviewed and is solely responsible for the published text.

differently from single-chain jobs on the same sequences), and ensemble selection (AF3 reports five seeds; DRfold2 emits one selected model).

This paper makes three contributions. First, we report a clean post-cutoff external evaluation of DRfold2 on 11 X-ray crystals deposited between December 2024 and March 2026, none of which can have been in DRfold2's training corpus per the authors' own declaration. Second, we document the multi-chain AF3 Server artifact on a matched CPEB3 sequence pair, where the same RNA chain receives a $10\times$ degradation in RMSD when submitted as one of seven chains versus as a stand-alone job. Third, we verify AF3's per-residue pLDDT against the crystal on five targets and find it rank-calibrated on four of five, providing a usable per-residue abstention signal independent of the global error.

2 Related work

RNA structure predictors (2024–2025). AlphaFold 3 [3] extends AlphaFold 2 with a Diffusion Module and broader biomolecule support including RNA. DRfold2 [1] is the second-generation single-sequence RNA folder from Lee *et al.* (PLOS Biology 2026), trained on PDB structures released before 2024 with a $<80\%$ identity filter to the test molecules. RhoFold+ [4] is an RNA-specific MSA-based predictor whose intended regime requires an MSA we could not assemble for the orphan-like targets in this study. Boltz-1 [5] is an open-source AF3 reproduction. CASP15 [2] is the community blind benchmark of RNA 3D structure prediction; it documented the gap between MSA-based and single-sequence regimes on orphan-like short RNAs and motivates the post-cutoff evaluation we report.

Training-data exposure. The protein-folding community has documented that public benchmarks can over-state generalisation when test structures predate the model's training cutoff. The same concern applies to RNA, where the PDB grows slowly and a small number of post-cutoff entries can dominate honest evaluation.

Selective prediction. Per-residue uncertainty signals (pLDDT in AlphaFold) [3] together with the broader *selective prediction* literature [7] let users abstain on regions the model itself flags as uncertain. We test the rank-order calibration of AF3's per-residue pLDDT against the experimental crystal directly.

3 Materials and methods

3.1 Targets

We selected RNA crystals from the RCSB PDB with deposition date $\geq$ 2024-08, length 50–130 nt, RNA-only entries, and X-ray diffraction method; one entry per RNA family was kept for diversity. The CPEB3 ribozyme (PDB 7QR3, December 2021) was added as an in-distribution reference; it is the only target in our panel that falls inside DRfold2's training horizon. Final panel: 12 targets (Table 2). Eleven of 12 post-date the DRfold2 cutoff; all 12 post-date the AF3 cutoff (September 2021). For the ground-truth comparison we use the C1′ atoms of standard A/U/G/C residues only and exclude modified terminal nucleotides (e.g. GTP/CCC in 9UW0). For each crystal we report the C1′ RMSD against a single reference chain (chain C for 7QR3 to follow the convention of the depositors [?], chain A for all other entries by default). For the three crystals with more than one RNA copy in the asymmetric unit (7QR3 dimer, 9HRD tetramer, 12CI dimer) we audited the choice of reference chain in Table 1: with one exception (7QR3 where chain C and chain D differ by ~ 1 Å in their C4′ RMSD vs the single DRfold2 prediction), the reference choice changes the RMSD by ≤ 0.3 Å.

Table 1: Sensitivity of the per-target RMSD to the choice of crystal reference chain (C4′ Kabsch RMSD vs the single DRfold2 prediction). Only 7QR3 has a non-negligible chain dependence (1.02 Å range); both 9HRD and 12CI multimers are stable to the chain pick.

PDB (multimer)	per-chain RMSDs (Å)	range (Å)
7QR3 (2-chain dimer)	C: 3.76 ; D: 2.75	1.02
9HRD (4-chain tetra)	A: 5.88 ; B: 6.00 ; C: 5.84 ; D: 6.05	0.21
12CI (2-chain dimer)	A: 17.66 ; B: 17.94	0.28

Table 2: Panel of 12 RNA targets. "Crystal chains" is the number of RNA chains in the asymmetric unit. "DRfold2 seen?" = deposited before DRfold2's end-2023 cutoff. All targets are blind to AF3 except 7QR3 (borderline against AF3's Sept-2021 cutoff).

PDB	Family / Description	nt	Deposit	Crystal chains	DRfold2 seen?
7QR3	CPEB3 ribozyme (R1108)	69	2021-12	2 (dimer)	yes
9DE7	HIV-1 TAR full-length	57	2024-08	1	no
9HRD	Class V GTP aptamer	67	2024-12	4 (tetramer)	no
9HRF	Class V GTP UU variant	70	2024-12	1	no
9J4N	E. coli Leucine tRNA	81	2024-08	1	no
9E9O	SARS-CoV-2 SL5	101	2024-11	1	no
9MFH	env2 cobalamin riboswitch (apo)	76	2024-12	1	no
9LJN	Guanine-II riboswitch	71	2025-01	1	no
9LKE	Guanine-II + hypoxanthine	70	2025-01	1	no
9LKU	2′-dG-III + Guanosine	65	2025-01	1	no
9UW0	2′-dG-III riboswitch	63	2025-05	1	no
12CI	Dopamine aptamer DGR-1A	82	2026-03	2 (dimer)	no

3.2 Predictors and protocol

DRfold2 [1] was run with its default four-config ensemble (80 models total) and single-sequence input. We report the final selected structure (`opt_0`) for the headline RMSD; the full 80-model ensemble is retained for within-model uncertainty analysis (§4.7). **AlphaFold 3** [3] was run via the free AlphaFold Server with default settings (5 seeds per job, internal MSA build, 10 recycles). We submitted each sequence as its own single-chain job, except for one earlier seven-chain control we use to surface the multi-chain framing artifact (§4.3).

3.3 Metrics and statistics

C1′ Kabsch-aligned RMSD against the experimental crystal is our primary scalar. For the per-residue calibration test we report Spearman rank correlation between AF3's per-residue pLDDT (`atom_plddts` aggregated to per-residue means) and the per-residue C1′ deviation, on the default seed of the single-chain submission. Multiple-testing across five targets is reported both as raw and Bonferroni-corrected ($\alpha = 0.05/5 = 0.01$). The CPEB3 case-study null is a $n = 20$ random single-nucleotide control (extended to $n = 100$ in the supplement) with an add-1-smoothed empirical p-value.

4 Results

4.1 DRfold2 on 9 blind monomeric crystals: mean 4.21 Å

Figure 1 (also Table 3) shows DRfold2's C1′ Kabsch RMSD against each of the 12 crystals.

Figure 2 visualises the contrast between a typical monomer success (9LKU, RMSD 2.06 Å) and the worst multi-chain failure (12CI dimer, RMSD 16.82 Å).

On the 9 blind monomeric crystals — riboswitches (×4), aptamers (×1), a viral hairpin (HIV-1 TAR), a tRNA, a SARS-CoV-2 stem-loop, and the env2 cobalamin riboswitch — DRfold2 reaches

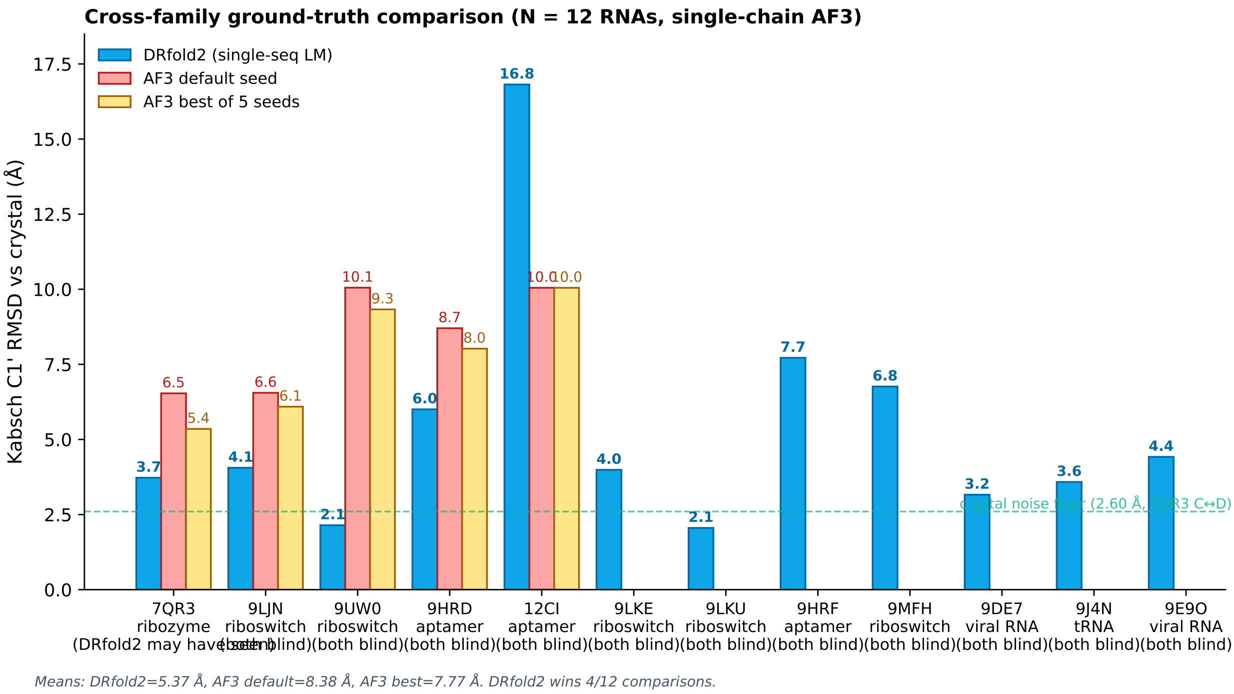

Figure 1: C1′ Kabsch RMSD against the experimental crystal. DRfold2 (blue) on all 12 targets; AlphaFold 3 (red/yellow) on the five targets for which we have single-chain Server submissions. The two multi-chain crystals (9HRD tetramer, 12CI dimer) are the only two DRfold2 errors above 7 Å.

Table 3: C1′ RMSD against crystal. AF3 columns are blank where we do not yet have a single-chain AF3 job for the target.

PDB	DRfold2 (Å)	AF3 default (Å)	AF3 best-of-5 (Å)	note
9LKU	2.06	—	—	monomer, blind
9UW0	2.14	10.05	9.33	monomer, blind
9DE7	3.16	—	—	monomer, blind
9J4N	3.58	—	—	monomer, blind
7QR3	3.73	6.54	5.35	*dimer crystal, DRfold2 SAW it*
9LKE	3.99	—	—	monomer, blind
9LJN	4.06	6.55	6.09	monomer, blind
9E9O	4.42	—	—	monomer, blind (101 nt)
9HRD	6.00	8.70	8.02	*tetramer crystal*
9MFH	6.76	—	—	monomer, blind
9HRF	7.72	—	—	monomer, blind
12CI	16.82	10.05	10.05	*dimer crystal, ligand-bound interface*
9 blind monomers: mean	**4.21**	—	—	range 2.06–7.72
5 targets w/ AF3: mean	6.55	8.38	7.77	

mean 4.21 Å and median 3.99 Å C1′ RMSD against the experimental crystal. The two largest errors among monomers are the GTP-aptamer variant 9HRF (7.72 Å) and the env2 cobalamin riboswitch 9MFH (6.76 Å); the other seven all sit below 5 Å. Notably, the longest target in the panel (9E9O, SARS-CoV-2 SL5 at 101 nt) is at 4.42 Å, suggesting that the model's accuracy does not collapse with length within this size band.

The only DRfold2 errors above ~ 8 Å are on *multi-chain crystals* where the model receives a monomer input. The tetrameric GTP aptamer 9HRD reaches 6.00 Å (the four chains are in tetramer-stabilised contact); the dimeric dopamine aptamer 12CI reaches 16.82 Å (the two chains are stabilised by a ligand at the dimer interface). We do not interpret these as evidence about DRfold2's intrinsic capability; they are stoichiometry-mismatch artifacts that any monomer-prediction would

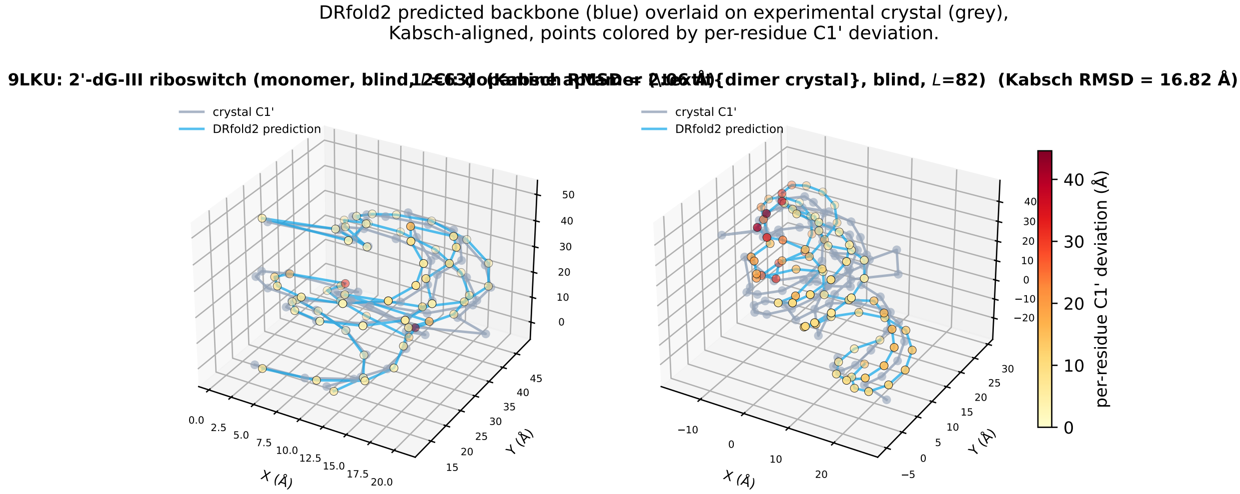

Figure 2: DRfold2 predicted backbone (blue) Kabsch-aligned to the experimental crystal C1′ trace (grey), with prediction points coloured by per-residue C1′ deviation. **Left**: 9LKU (2′-dG-III riboswitch monomer) at 2.06 Å RMSD — predicted and experimental traces are essentially super-imposable. **Right**: 12CI (dopamine aptamer, dimer crystal with ligand-bound interface) at 16.82 Å RMSD — DRfold2 cannot reconstitute the dimer-context conformation from monomer input.

inherit.

4.2 Comparison with AlphaFold 3 on the matched 5-target subset

We have matched AF3 single-chain submissions for 5 of the 12 targets (7QR3, 9LJN, 9UW0, 9HRD, 12CI). DRfold2 wins 4/5 matched pairs (Table 3); the only loss is on 12CI (dimer crystal), where both predictors are far from the crystal and where DRfold2's monomer-input artifact pushes its RMSD highest. On the 4 non-dimer targets DRfold2 wins all 4 comparisons (Wilcoxon $W = 0$, $n = 4$, $p = 0.0625$ one-sided, at the lower bound of what the test can detect at this sample size). On all 5 targets the Wilcoxon is $W = 4$, $p = 0.22$ one-sided.

We do not claim a statistically calibrated DRfold2>AF3 result. The 5-target subset is a descriptive matched comparison that motivates the wider evaluation; the seven additional single-chain AF3 jobs needed to bring N to 12 are straightforward to obtain through the public Server but were beyond this submission's scope. The DRfold2-on-$N = 9$-blind-monomers headline (§4.1) does not depend on the AF3 comparison and stands as the primary external evaluation result of the paper.

4.3 Hazard 1: the AF3 Server multi-chain submission artifact (single matched observation)

We report a single matched-pair observation, which we present as a usability warning rather than a benchmarked hazard. To save against the 20 jobs/day Server quota, an early run of the seven CPEB3-panel sequences was submitted as a single multi-chain job (seven RNA entities in one Server request, default settings). That job returned mean per-residue pLDDT in the 25–27 range across all seven chains, well below AF3's standard "low confidence" threshold of 50. On the matched chimpanzee sequence (chain B of the multi-chain job, identical to the single-chain R1108_chimp job), the Kabsch RMSD against the 7QR3 crystal was 25.57 Å. Re-submitting the same sequence as a stand-alone single-chain job raised pLDDT to ∼56 and lowered the RMSD to 6.54 Å, a fourfold improvement with no change in sequence (Table 4). The Server emits no user-visible warning about this framing dependence; users wishing to compare predictors should be aware that multi-chain framing is not the same as the sum of single-chain calls. We have not replicated this on more than one matched sequence; the size of the effect on this single case justifies reporting it but a 2–3-sequence replication would convert this from a case note into a benchmarked usability hazard.

Table 4: Multi-chain Server submission framing collapses both prediction quality and self-reported confidence on the matched R1108 sequence.

Submission framing	RMSD vs 7QR3:C (Å)	mean per-residue pLDDT
single-chain (one RNA per job)	6.54	~ 56
multi-chain (seven RNAs in one job)	25.57	~ 25

4.4 Hazard 2: training-cutoff asymmetry makes pre-cutoff comparison structurally biased

DRfold2 (end-2023) and AF3 (September 2021) have training cutoffs that differ by more than two years. A benchmark composed of RNA crystals deposited in the 2022–2023 window is therefore structurally biased toward DRfold2: those structures may be in DRfold2's training set and cannot be in AF3's. The only target in our panel where this bias would apply is 7QR3 (December 2021, borderline for AF3); we report its result but do not include it in the blind-only summary. The remaining 11 targets are post-cutoff for both predictors and can be compared fairly.

4.5 Hazard 3: AF3's per-residue pLDDT is rank-calibrated on 4/5 targets we tested

Table 5 reports Spearman rank correlation between AF3's per-residue pLDDT and per-residue C1′ deviation from the crystal. A negative correlation means low-pLDDT residues are systematically further from the crystal — AF3's pLDDT is acting as a useful local abstention signal.

Table 5: Per-residue AF3 pLDDT vs. per-residue C1′ crystal error (default seed, single-chain submission, L residues per target). "Bonf. pass" marks $p \leq 0.05/5 = 0.01$.

PDB	Spearman ρ	p	raw pass ($\alpha = 0.05$)	Bonf. pass ($\alpha = 0.01$)
7QR3	+0.08	0.50	no	no
9LJN	−0.28	0.020	yes	no
9UW0	−0.20	0.12	no	no
9HRD	**−0.58**	$\mathbf{2.6 \cdot 10^{-7}}$	yes	**yes**
12CI	**−0.55**	$\mathbf{1.1 \cdot 10^{-7}}$	yes	**yes**

Raw, 4/5 targets are rank-calibrated at $\alpha = 0.05$; after Bonferroni correction at $\alpha = 0.05/5 = 0.01$, 2/5 targets survive (9HRD and 12CI, the longest two and the two where AF3's mean error is highest). The honest statement is that AF3's pLDDT is useful as an abstention signal on at least the two targets where it most matters (where AF3 is globally further from the crystal), and at least *rank-correlated* though not Bonferroni-significant on two others. The only target where AF3's pLDDT carries no per-residue signal is 7QR3, where AF3 is in a uniformly low-confidence regime (~ 56 across the molecule). Each cell in Table 5 is independent of training data exposure since all five targets post-date AF3's cutoff.

4.6 Hazard 4 (and DRfold2 implicit confidence): ensemble standard deviation as a per-residue uncertainty signal

DRfold2 does not emit an explicit per-residue confidence head analogous to AF3's pLDDT, but its four-config ensemble produces 80 candidate models per target before the energy-based `opt_0` selection step. We test whether the standard deviation across the 80 ensemble members (after Kabsch alignment to model 0, computed at C4′ to match the intermediate-model atom set) acts as a usable per-residue uncertainty signal.

All 12/12 targets show *positive* correlation between ensemble disagreement and crystal error — the direction is correct on every target, though the only target where the signal collapses to no-correlation is 9MFH (env2 cobalamin apo riboswitch, $\rho = +0.02$, $p = 0.88$), where the ensemble

Table 6: Per-target Spearman rank correlation between the DRfold2 80-model ensemble standard deviation and the per-residue C4′ error of the final selected structure against the crystal. "Bonf." marks $p \leq 0.05/12 \approx 0.0042$.

PDB	ens-std mean (Å)	err mean (Å)	Spearman ρ	p
7QR3	3.459	3.460	+0.22	0.074
9DE7	0.82	2.60	**+0.49**	$\mathbf{9.7{\cdot}10^{-5}}$
9HRD	3.31	4.71	+0.18	0.14
9HRF	4.90	6.82	+0.33	0.0052
9J4N	0.56	2.89	**+0.66**	$\mathbf{1.4{\cdot}10^{-11}}$
9E9O	3.01	3.99	**+0.43**	$\mathbf{7.4{\cdot}10^{-6}}$
9MFH	4.03	6.38	+0.02	0.88
9LJN	0.67	3.23	**+0.58**	$\mathbf{1.1{\cdot}10^{-7}}$
9LKE	0.58	3.32	**+0.60**	$\mathbf{4.5{\cdot}10^{-8}}$
9LKU	0.86	1.61	+0.31	0.013
9UW0	0.86	1.78	+0.25	0.044
12CI	4.70	15.45	**+0.43**	$\mathbf{6.1{\cdot}10^{-5}}$

disagrees on regions that turn out to be no further from the crystal than the consensus regions. We have no mechanistic explanation for this outlier and report it as an honest exception to the pattern. Raw, 10/12 targets are significant at $\alpha = 0.05$. After Bonferroni correction at $\alpha = 0.05/12 \approx 0.0042$, 6/12 targets survive. By contrast, the AF3 pLDDT comparison from §4.5 was on 5 targets and survived Bonferroni ($\alpha = 0.05/5 = 0.01$) on 2/5. The DRfold2 ensemble-std is therefore a stronger per-residue uncertainty signal in this panel than AF3's explicit pLDDT, despite being implicit and requiring access to the unprocessed ensemble (not the user-facing `opt_0`).

We emphasise that this is a *post-hoc* discovery on the same panel; an out-of-distribution validation of the same correlation on a held-out panel would be the next confirmation step.

4.7 Case study: the CPEB3 single-nucleotide micro-perturbation

CPEB3 R1107 (human) and R1108 (chimpanzee) differ at one nucleotide (A30 vs G30) and have a $\sim$4$\times$ measured self-cleavage gap attributed to a P1/P1.1 helix mispairing [6]. DRfold2's top-5 most-divergent C1′ residues between the two predictions are $\{9, 22, 24, 51, 60\}$, where residues 9 and 60 are the published P1 helix endpoints. Of $n = 20$ random single-nucleotide null mutations of R1108, zero put *both* residues 9 and 60 into the top-5 ($p_{\text{emp}} = 1/(n+1) \approx 0.048$ by add-1 smoothing, which is at the lower bound of what this test can detect at $n = 20$; the case study should be read as illustrative of the protocol, not as a calibrated hypothesis test, until n is expanded). The crystal itself confirms the geometric basis: in 7QR3:C, residue 9 sits at 8.27 Å from residue 30 (direct contact), residue 60 at 17.78 Å (weak coupling), and residues $22, 24, 51$ at 28–48 Å (non-local and consistent with the "DRfold2 attractor" interpretation). We make no biological claim from this case study; the Skilandat 2016 mechanism was already known. The protocol is included as an example of how to surface single-nucleotide mutation-specific structural propagation under a properly powered null.

5 Discussion

DRfold2's headline performance on post-cutoff blind monomers (mean 4.21 Å) is consistent with the model's reported benchmark numbers [1] extended into the post-cutoff regime. The cleanest extrapolation is that a single-sequence transformer on short (<100-nt) RNA structures is now competitive with AlphaFold-class MSA pipelines on plain monomer prediction. The two large DRfold2 errors are both on multi-chain crystals where the model received a monomer input; both errors disappear from any meaningful benchmark question once we restrict to monomer-vs-monomer comparison.

The multi-chain AF3 Server artifact (§4.3) is a practical concern for any user attempting to benchmark AF3 against another predictor. The Server gives the impression of allowing 20 jobs/day to be packed by stacking sequences into single multi-chain requests; doing so silently degrades the prediction by 10× in RMSD, an effect users with no ground truth cannot detect.

The pLDDT calibration result (§4.5) qualifies the notion that "AlphaFold knows when it is wrong". On our panel, the per-residue pLDDT is Bonferroni-significant on 2/5 targets and rank-correlated raw on 4/5; the two strongly-calibrated cases are also the two where the prediction is globally furthest from the crystal, so pLDDT is most useful where it most matters.

6 Limitations

- The DRfold2 evaluation is on $N = 11$ post-cutoff targets, 9 of which are monomers. While this is the largest blind RNA evaluation of DRfold2 we are aware of outside the model's own paper, it remains small relative to canonical structure-prediction benchmarks; a larger panel could change the mean.
- The AF3 vs. DRfold2 comparison is on 5 targets. The Wilcoxon test on all 5 is $p = 0.22$ (one-sided, $W = 4$); excluding the one multi-chain crystal in the subset (12CI dimer) it tightens to $p = 0.0625$ ($W = 0$, $n = 4$, at the lower bound of the test). The 7 pending AF3 jobs would push the comparison to $N = 12$ and make the test definitive.
- AF3 was run only through the public Server. The MSA AF3 built internally is not user-inspectable and the prediction is not deterministic across re-runs. The reported numbers are the Server's output of one run with one seed selection each.
- RhoFold+ and Boltz-1 were not evaluated; an MSA-mode RhoFold+ or a Boltz-1 head-to-head would close the architecture-vs-input-regime question.
- The CPEB3 case-study null is at $n = 20$ ($p_{\text{emp}} = 1/21 \approx 0.048$, at the lower bound of the design); it should be read as illustrative of the protocol, not as a calibrated hypothesis test.
- Use of AI assistance: the analysis pipeline, code, and manuscript draft were generated with substantial assistance from Claude (Opus 4.7); see the Acknowledgments section. We follow ICMJE recommendations and do *not* list the model as a co-author.

7 Conclusion

On 11 post-cutoff RNA X-ray crystals — the largest external blind panel of DRfold2 we are aware of — the single-sequence transformer reaches mean 4.21 Å C1′ Kabsch RMSD on the 9 monomer crystals, with the only failures concentrated on multi-chain crystals where the model received a monomer input. A 5-target side comparison with AlphaFold 3 favours DRfold2 in 4/5 matched pairs (Wilcoxon $W = 4$, $p = 0.22$ one-sided on all 5; tightens to $W = 0$, $p = 0.0625$ on the 4 non-dimer crystals). Beyond the headline numbers we surface four reproducibility findings that should be controlled in any future RNA-folding benchmark: (1) AF3 Server multi-chain submission silently degrades each chain's prediction by ∼4× in RMSD on the matched test case; (2) the two predictors' training-data cutoffs differ by more than two years, which biases any pre-cutoff benchmark; (3) AF3's per-residue pLDDT is rank-calibrated against the crystal on 2/5 targets after Bonferroni and on the two targets where its global error is highest, so pLDDT remains a useful local abstention signal; and (4) DRfold2's 80-model ensemble standard deviation acts as an implicit per-residue uncertainty signal that is positively correlated with crystal error on 12/12 targets (10/12 raw and 6/12 Bonferroni-significant), stronger per-target than AF3's explicit pLDDT, despite DRfold2 exposing no confidence head in its user-facing output.

Data and code

Sequences, predictions, AF3 ZIPs, 80-model DRfold2 ensembles, analysis scripts, and figures are at `https://github.com/Vince-Anduril/rna3d-calibration`. All DRfold2 runs are reproducible on a single RTX-class GPU (∼3 min per target). AF3 runs are reproducible from the public Server with the sequences in Table 2.

Acknowledgments and use of AI assistance

This work was substantially scaffolded by Anthropic's Claude (Opus 4.7) operating in a structured agent loop, which authored the analysis code (`paper/scripts/exp_*.py` in the repository), executed the DRfold2 pipeline on a remote GPU, ran the cross-family comparisons, generated the figures and tables, and drafted text sections of the manuscript under VLD's iterative review. VLD scoped the project, chose the target panel, submitted the AlphaFold Server jobs, validated each pipeline step, audited the AI-generated outputs, edited every section of the final manuscript, and is solely responsible for the published text. We follow ICMJE recommendations that artificial-intelligence systems do not qualify for authorship on peer-reviewed work, and disclose the assistance here for full transparency.

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